

Weibull distribution A continuous random variable X is said to follow a weibull distribution with parameters $\alpha, \beta > 0$ if X has the probability density function

$$f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0. \\ 0 & \text{otherwise.} \end{cases}$$

Note:- If $\beta = 1$ then X follows exponential distribution with parameter α .

r^{th} moment of the weibull distribution

$$E(x^r) = \int_{-\infty}^{\infty} x^r f(x) dx = \int_0^{\infty} x^r \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} dx$$

$$= \alpha \beta \int_0^{\infty} x^{r+\beta-1} e^{-\alpha x^\beta} dx$$

$$\int_0^{\infty} \left(\frac{y}{\alpha}\right)^{\frac{r}{\beta}} e^{-y} dy$$

put $y = \alpha x^\beta$ then $dy = \alpha \beta x^{\beta-1} dx \therefore dx = \frac{dy}{\alpha \beta x^{\beta-1}}$

$$\therefore E(x^r) = \alpha \beta \int_0^{\infty} \left[\left(\frac{y}{\alpha}\right)^{\frac{1}{\beta}}\right]^{r+\beta-1} e^{-y} \frac{dy}{\alpha \beta \left[\left(\frac{y}{\alpha}\right)^{\frac{1}{\beta}}\right]^{\beta-1}}$$

$$= \int_0^{\infty} \left(\frac{y}{\alpha}\right)^{\frac{r}{\beta} + 1 - \frac{1}{\beta}} e^{-y} \left(\frac{y}{\alpha}\right)^{\frac{1}{\beta}-1} dy$$

$$= \alpha^{-\frac{r}{\beta}} \int_0^{\infty} y^{\frac{r}{\beta}} e^{-y} dy$$

$$E(x^r) = \alpha^{-\frac{r}{\beta}} \Gamma\left(\frac{r}{\beta} + 1\right)$$

Mean and Variance

$$E(X) = \text{Mean} = \alpha^{-\frac{1}{\beta}} \Gamma\left(\frac{1}{\beta} + 1\right)$$

$$E(X^2) = \alpha^{-2/\beta} \Gamma\left(\frac{2}{\beta} + 1\right)$$

$$\begin{aligned} \text{Variance} &= E(X^2) - [E(X)]^2 = \alpha^{-2/\beta} \Gamma\left(\frac{2}{\beta} + 1\right) - \left[\alpha^{-\frac{1}{\beta}} \Gamma\left(\frac{1}{\beta} + 1\right) \right]^2 \\ &= \alpha^{-2/\beta} \left[\Gamma\left(\frac{2}{\beta} + 1\right) - \left[\Gamma\left(\frac{1}{\beta} + 1\right) \right]^2 \right] \end{aligned}$$

Prob Each of the 6 tubes of a radio set has a life length (in years) which may be considered as a random variable that follows a weibull distribution with parameters $\alpha = 25$ and $\beta = 2$. If these tubes function independently of one another what is the probability that no tube will have to be replaced during the first 2 months of service.

Soln:-

Let X represents the life length of each tube (in years)
Then its density function is

$$f(x) = \alpha \beta x^{\beta-1} e^{-\alpha x^{\beta}}; \quad x > 0$$
$$= 50 x e^{-25x^2}; \quad x > 0$$

$P(\text{a tube is not to be replaced during the first 2 months})$

$$= P\left(X > \frac{1}{6}\right) = \int_{\frac{1}{6}}^{\infty} 50 x e^{-25x^2} dx$$

$$= \int_{\frac{1}{6}}^{\infty} d(-e^{-25x^2}) = \left[-e^{-25x^2}\right]_{\frac{1}{6}}^{\infty}$$

$$= e^{-25/36}$$

P (all the 6 tubes are not replaced during first

$$2 \text{ months}) = \left(e^{-25/36} \right)^6$$

$$= 0.0155.$$

(by independence) $\left(\begin{array}{l} P(A \cap B) \\ = P(A)P(B) \end{array} \right)$